

translating the plant's signals into traditional electronics.

Creating circuits from plant parts

The manufacturing of analogue and digital organic electronic circuits and devices in living plants has been reported. The four key components of a circuit (an energy source, a conductor (wire), an electrical load, and at least one controller) have been achieved using the xylem, leaves, veins, and signals of the plant as the pattern and integral part of the circuit elements and functions.

With the help of integrated and distributed electronics in plants, we can imagine a range of applications including energy harvesting from photosynthesis, precision recording and regulation of plant physiology, and alternatives to genetic modification for plant optimisation.

Where research stands today

Bioelectronic devices for plant monitoring and optimisation are the desired tools for plant biologists, agriculture, and forestry. Bioelectronic devices, sensors, and actuators based on organic electronic and iontronic materials for plant interface are being designed and developed.

Creation of plant bioelectronic devices will define new means for decoding and manipulating plant biology from the cellular level to the organism level. These devices will offer unique characteristics including dynamic control of physiology of plant and signal monitoring with high spatiotemporal resolution. These devices will be compatible with both wild type and genetically engineered plants.

The goal is to develop technologies with a focus on understanding and enhancing plant responses to environmental stress. Efforts to introduce electronics in plants for the paper industry originated with the idea of printed electronics on paper.

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From a rose to transistor

Researchers of a university in Sweden have created analogue and digital

electronic circuits inside living plants by using the vascular system of living roses to build key components of electronic circuits. They have successfully demonstrated that wires, digital logic, and even display elements, when fabricated inside plants, can generate new applications for organic electronics and new tools in plant science.

The research team tried to introduce conductive polymers through rose stems and only one polymer (called PEDOT-S) successfully assembled itself inside the xylem channels as conducting wires, while still allowing the transport of water and nutrients inside the plant. They have used the material to create long (10cm) wires in the xylem channels of the rose.

Researchers were able to create an electrochemical transistor (a transistor that converts ionic signals to electronic output) by combining the wires with the electrolyte that surrounds these channels. Further, digital logic gate function was also demonstrated using the xylem transistors.

Scientists used methods common in plant biology like vacuum infiltration to infuse another PEDOT variant into the leaves; then the infused polymer formed pixels of electrochemical cells partitioned by the veins. The applied voltage caused the polymer to interact with the ions in the leaf, subsequently changing the colour of the PEDOT in a display-like device. This functioning is similar to that of roll-printed displays. However, these results are simply early steps to merge the diverse fields of organic electronics and plant science, but may lead to possible electronics plants in the near future.

Plant bioelectronics as a low-power sensor

With the growing use of electronic devices in our daily lives, scientists and engineers are also exploring ways to integrate plants' electronic functionality with human and electronic devices. A research team has built a device that can run on available power from a plant. The circuit built with parts measuring about 130nm consumes just 10nW of power, on average, during operation.

Normal electronics is not going to run on the types of currents and

voltages obtained from a tree as the nanoscale is not just in size but also in the energy and power consumption. Electric power to be extracted from plants, that is 'Tree power,' is unlikely to replace solar power for most applications, but the electronic output from plant bioelectronics system could provide a low-cost option for powering tree sensors that might be used to gauge a tree's health to detect environmental conditions or forest fires.

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Recently, uncontrolled and devastating forest fires in the USA have forced researchers to think critically about bioelectronics, that is, creating electronic plants. A researchers' team at MIT (USA) is investigating whether energy generated from trees can power a network of sensors to prevent spreading of forest fires. This team also claims that the technology will open the possibility of using trees as silent sentinels along the nation's borders to detect smuggled radioactive materials, as well as provide better local climate data towards fire prediction.

In-vivo circuits paving the way for electronic plants

Creating electronic plants can become a reality with a new technique that creates in-vivo electronic circuits. Controlling and interfacing with chemical pathways inside the plants could pave the way to various exotic applications as photosynthesis-based fuel cells, sensors, and growth regulators and devices that will be helpful to modulate the internal functions of plants.

New ways of interacting with electronic plants will help to develop applications for energy and environmental sustainability. Sensors employed in plants that make use of the energy formed in the chlorophyll will produce green antennas or produce new materials. **EFY**



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